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### Theoretical Trajectory 2

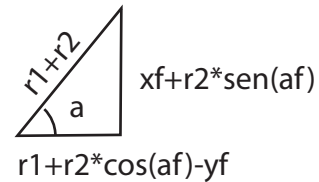
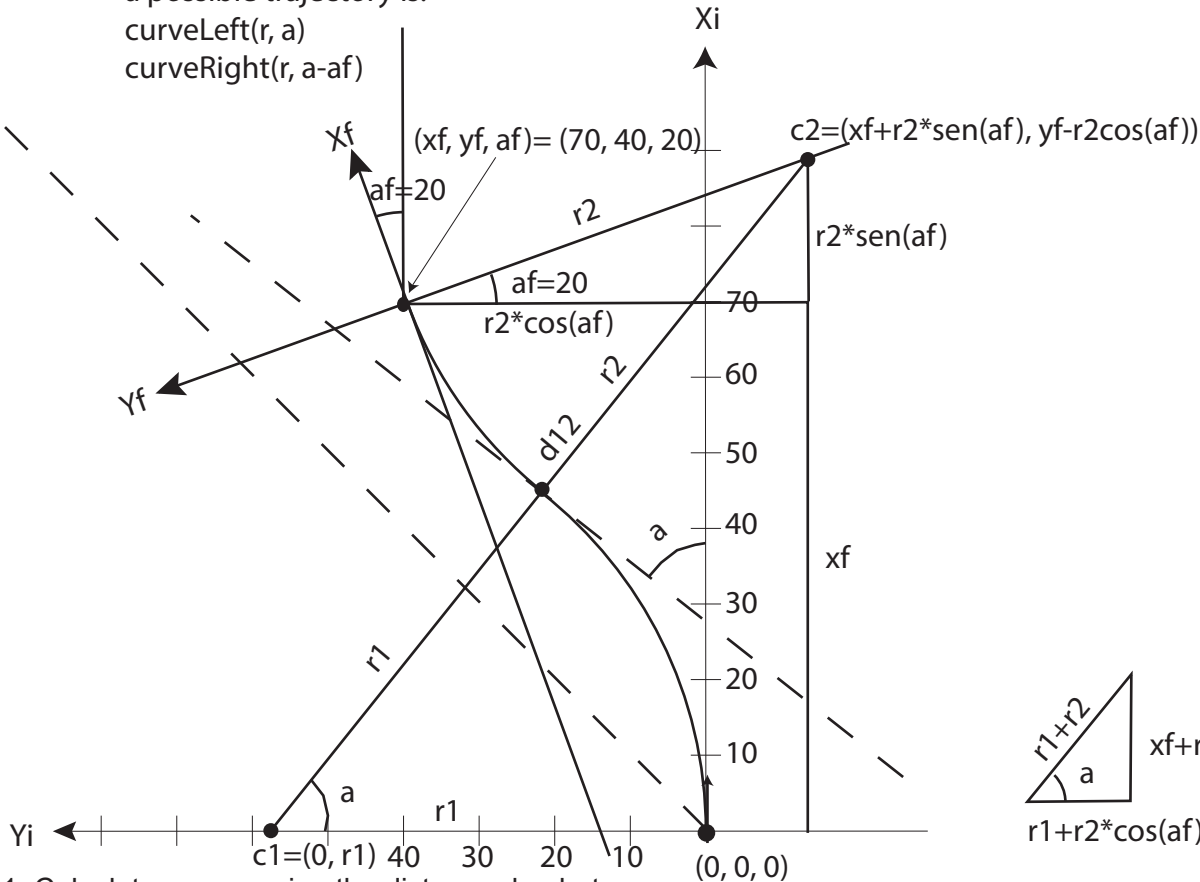
Consider a final point  $(x_f, y_f, a_f)$ .

When  $x_f > y_f$  and  $45 \geq a_f > -90$  and  $y_{c2} < y_{c1}$ ,

a possible trajectory is:

curveLeft( $r, a$ )

curveRight( $r, a - a_f$ )



1. Calculate  $r_1$  e  $r_2$  using the distance  $d_{12}$  between points  $c_1$  and  $c_2$ , which define the center points of two circumference arcs of each curve:

$$c_1 = (x_{c1}, y_{c1}),$$

$$x_{c1} = 0 \text{ and } y_{c1} = r_1, \text{ implies}$$

$$c_1 = (0, r_1).$$

and,

$$c_2 = (x_{c2}, y_{c2}),$$

$$x_{c2} = x_f + r_2 \cdot \sin(a_f) \text{ and } y_{c2} = y_f - r_2 \cdot \cos(a_f), \text{ implies}$$

$$c_2 = (x_f + r_2 \cdot \sin(a_f), y_f - r_2 \cdot \cos(a_f)).$$

then,

$$d_{12} = ((x_f + r_2 \cdot \sin(a_f))^2 + (r_1 - y_f + r_2 \cdot \cos(a_f))^2)^{0,5} = r_1 + r_2.$$

As the equation of  $d_{12}$  has two unknown variables  $r_1$  e  $r_2$ , there are three solutions to get only one unknown variable:

- i.  $r_1 = r_2 = r \Rightarrow (2 \cdot r)^2 = (x_f + r \cdot \sin(a_f))^2 + (r \cdot (\cos(a_f) + 1) - y_f)^2$ .
- ii.  $r_1 = C \Rightarrow (r_2 + C)^2 = (x_f + r_2 \cdot \sin(a_f))^2 + (C + r_2 \cdot \cos(a_f) - y_f)^2$ .
- iii.  $r_2 = C \Rightarrow (r_1 + C)^2 = (x_f + C \cdot \sin(a_f))^2 + (r_1 + C \cdot \cos(a_f) - y_f)^2$ .

2. Calculate angle  $a$ ,  
 $\sin(a) = (x_f + r_2 \cdot \sin(a_f)) / (r_1 + r_2)$ ,  
 $a = \arcsin((x_f + r_2 \cdot \sin(a_f)) / (r_1 + r_2))$ ,
3. For the example (70, 40, 20) and considering solution i., the terms of resolvent formula are,  
 $a = 2 - 2 \cdot \cos(a_f)$ ,  
 $b = 2 \cdot y_f \cdot (1 + \cos(a_f)) - 2 \cdot x_f \cdot \sin(a_f)$ ,  
 $c = -(x_f^2 + y_f^2)$ ,  
applying resolvent formula,  
 $r = 56,95 \Rightarrow r_1 = r_2 = 56,95$   
from a equation of point 2, will get  
 $a = 51,78$ .

The theorethical trajetory is,  
CurveLeft(56,95, 51,78)  
CurveRight(56,95, 31,78).